

Supplementary Material

Interactive Pattern Mining using Discriminant Sub-patterns as Dynamic Features

Arnold Hien^{1,2} , Samir Loudni² , Nouredine Aribi³ , Abdelkader Ouali¹ , and
Albrecht Zimmermann¹ 

¹ Normandie Univ., UNICAEN, CNRS – UMR GREYC, France

{abdelkader.ouali, albrecht.zimmermann}@unicaen.fr

² TASC – DAPI, IMT-Atlantique, LS2N – CNRS, 4 rue Alfred Kastler, 44307 Nantes, France

{samir.loudni, arnold.hien}@imt-atlantique.fr

³ Lab. LITIO, University of Oran1, 31000 Oran, Algeria

aribi.nouredine@univ-oran1.dz

1 Function SamplePatterns

Algorithm 1: Function SAMPLEPATTERNS

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1 In: Dataset  $\mathcal{D}$ , Sampling weight logistic function  $\varphi_{logistic} : \mathcal{L} \rightarrow [A, 1]$   
2 Parameters: Weight vector  $w$ , range  $A$ , FLEXICS: error tolerance  $\kappa$   
3 Out: TOP( $m$ ):  $m$  highest-weighted patterns;  
4 Function SamplePatterns ( $\mathcal{D}, \varphi_{logistic}$ )  
5    $C = \text{SAMPLERANDOMCELLWITHFLEXICS}(\mathcal{D}, freq(\cdot) \geq \theta, w, \kappa)$   
6   return TOP(1) pattern with the highest quality  $\varphi_{logistic}$  from  $C$ 
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Any sampling technique can be used to sample patterns in our approach [1,3,4]. In our experiments, to select the k patterns to show to the user, we use FLEXICS [3] to draw k weighted random samples proportional to $\varphi_{logistic}$. The main idea behind FLEXICS consists of partitioning the search space into a number of cells, and then sampling a solution from a random cell. This technique was inspired by recent advances in weighted solution sampling for SAT problems [5]. It exploits WEIGHTGEN [2] to sample SAT problem solutions. WEIGHTGEN partitions the search space into cells using random XOR constraints and then extracts a pattern from a randomly selected cell. Together, m XOR constraints identify one cell belonging to a partitioning of the overall solution space into 2^m cells. WEIGHTGEN uses a two-step procedure consisting of 1) estimating the number of XOR required for the partitioning and 2) generating these random constraints and sample the solutions. The sampling step needs the use of an efficient oracle to enumerate the solutions in the different cells. We used EFLEXICS which employs the ECLAT Algorithm for the enumeration. Algorithm 1 shows the pseudo-code of the function SamplePatterns. First, it samples a random cell, and then picks from that cell the TOP(1) pattern with the highest quality $\varphi_{logistic}$.

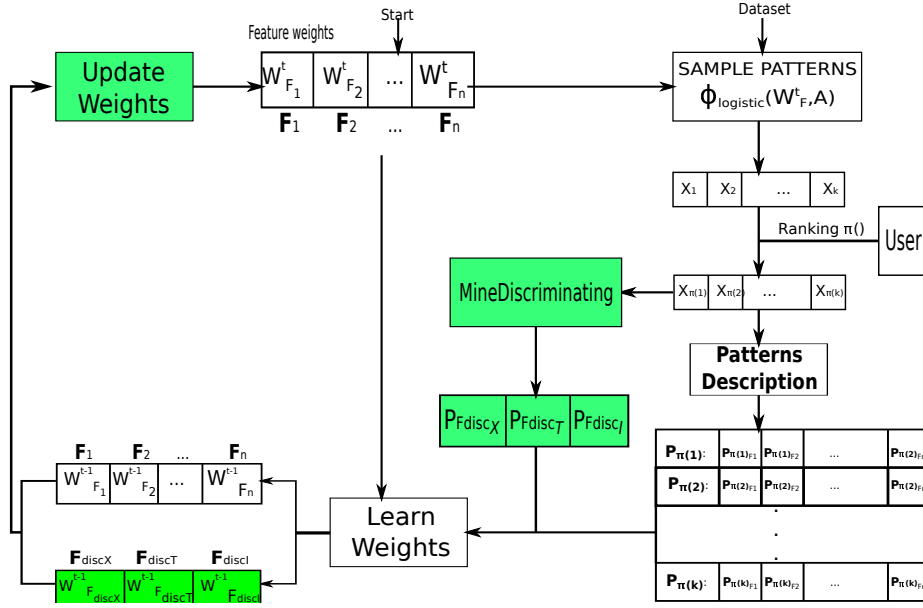


Fig. 1: Discriminating Sub-Patterns based learning schema of DISPALE.

2 Learning weights for $\varphi_{logistic}$ from discriminating sub-patterns

Figure 1 gives a detailed view of our new learning schema.

3 Complementary results for cumulative and non-cumulative $Regret_{max}$ values

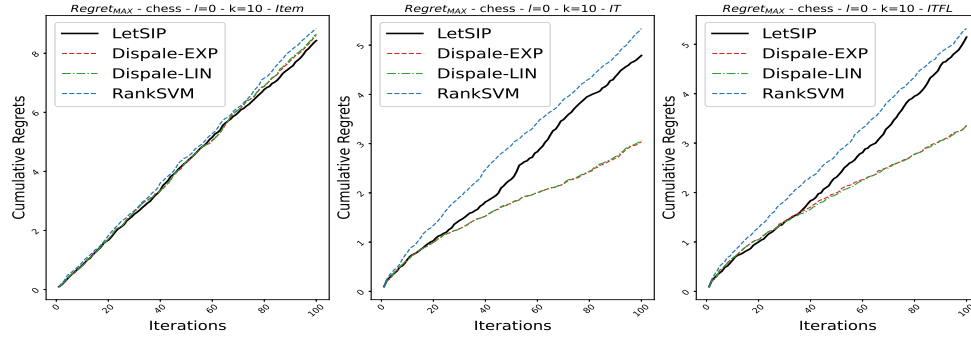
Figures 2-5 plot a detailed view of comparison between DISPALE, LETSIP and RANKSVM (cumulative and non-cumulative regret) for different datasets w.r.t. *maximal* quality, $k = 10$ and $\ell = 0$.

4 Complementary results for $Regret$ values

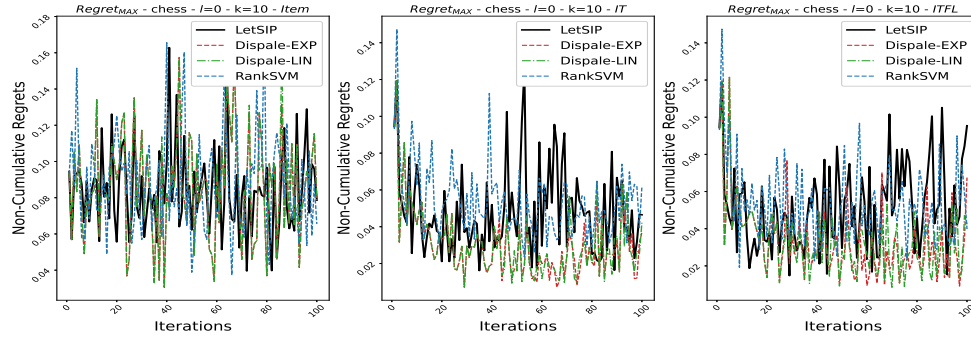
Tables 1-3 give detailed values of the regret for each dataset. Each table reports results for each combination of features : I $\sim$ Items, IT $\sim$ Items and Transactions, ITLF $\sim$ Items, Transactions, Length and Frequency.

5 Complementary results for CPU-times comparison

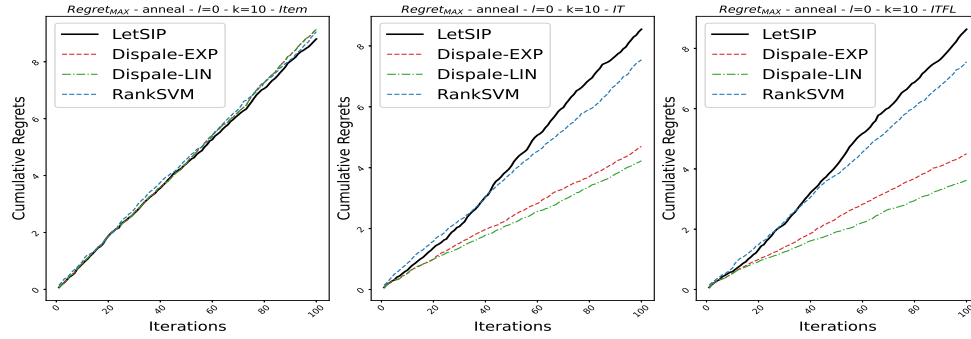
Figure 6 compares the CPU-times of LETSIP, DISPALE and RANKSVM w.r.t. the ILFT feature combination with $k = 10$ and $\ell = 0$.



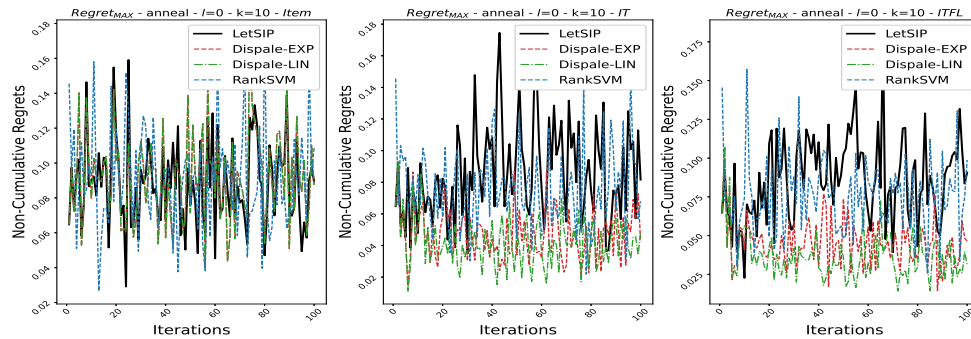
(a) Chess: cumulative regret.



(b) Chess: non cumulative regret.

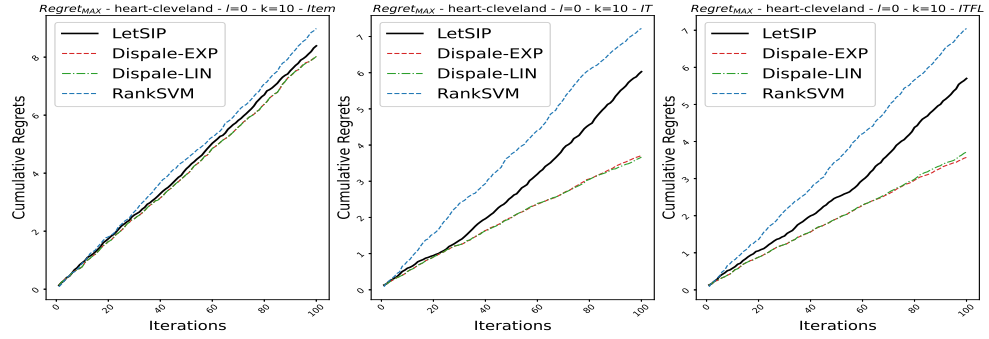


(c) Anneal: cumulative regret.

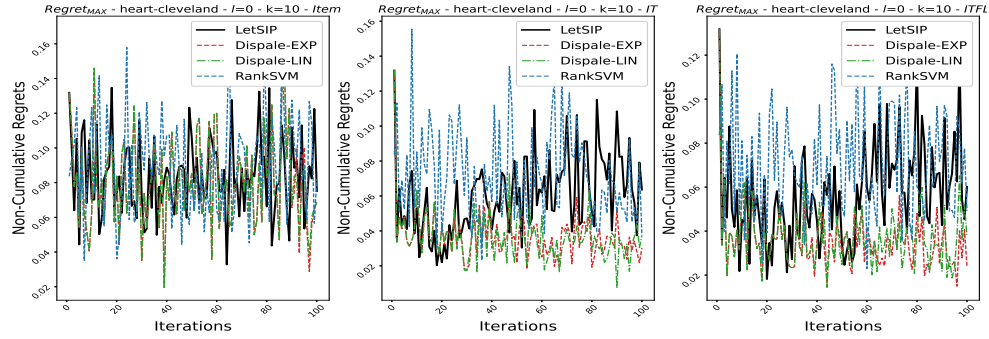


(d) Anneal: non cumulative regret.

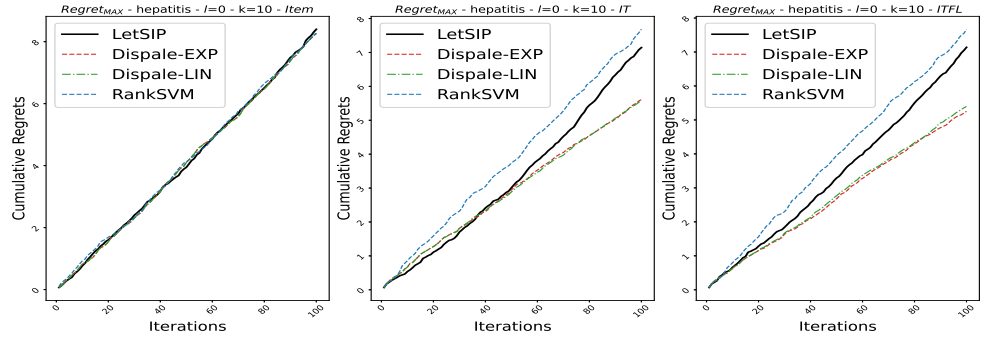
Fig. 2: A detailed view of experimental results w.r.t. $Regret_{\max}$, $k = 10$ and $\ell = 0$.



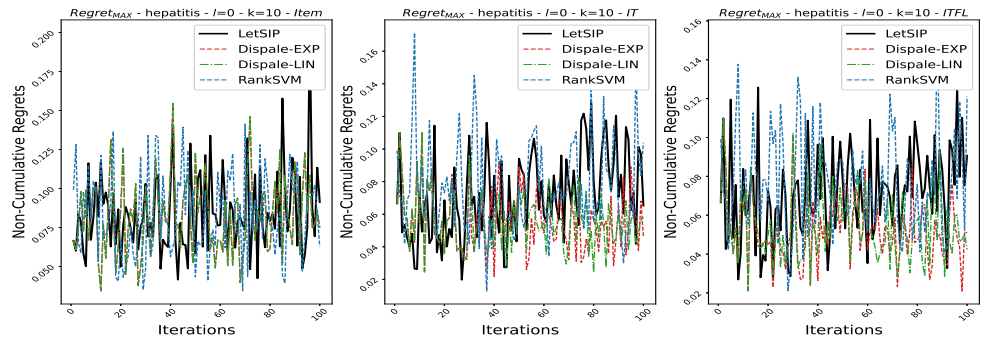
(a) Heart-cleveland: cumulative regret.



(b) Heart-cleveland: non cumulative regret.

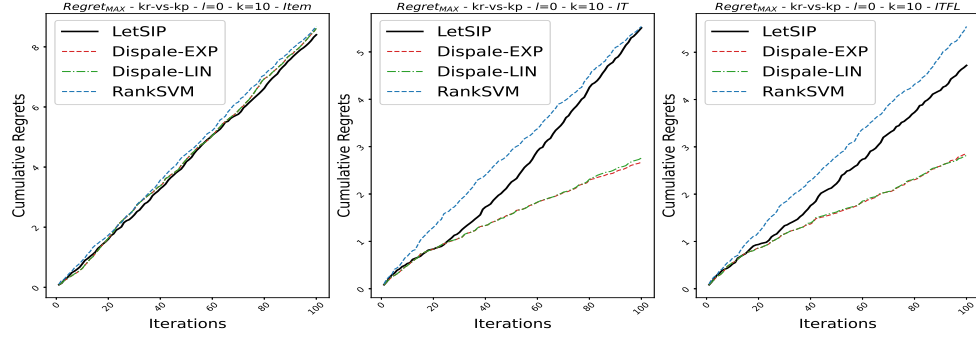


(c) Hepatitis: cumulative regret.

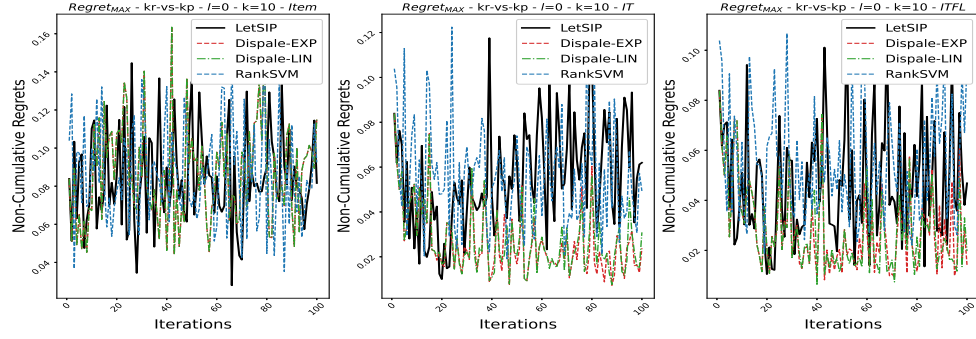


(d) Hepatitis: non cumulative regret.

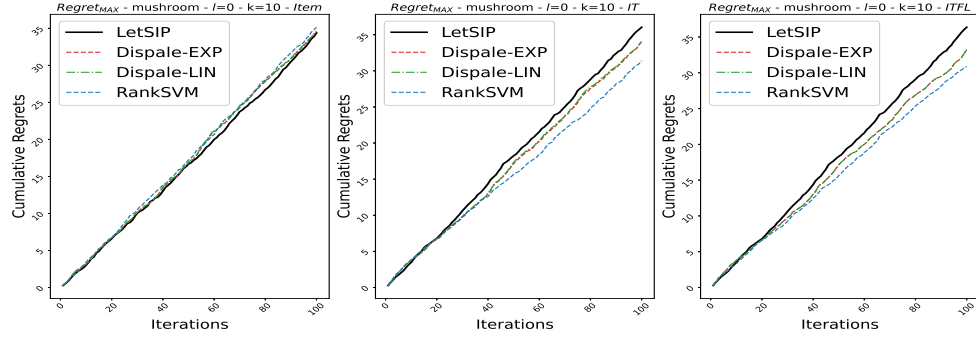
Fig. 3: A detailed view of experimental results w.r.t. $Regret_{\max}$, $k = 10$ and $\ell = 0$.



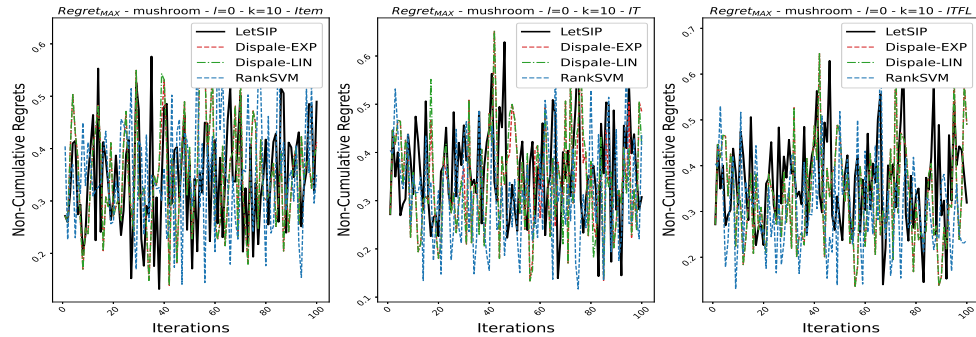
(a) Kr-vs-kp: cumulative regret.



(b) Kr-vs-kp: non cumulative regret.

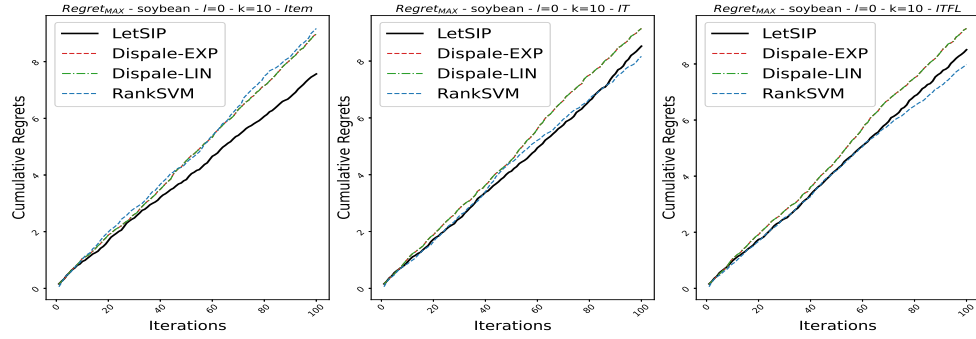


(c) Mushroom: cumulative regret.

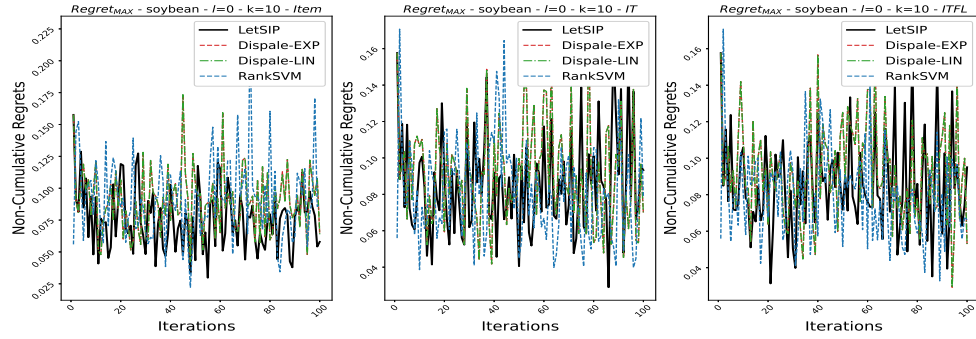


(d) Mushroom: non cumulative regret.

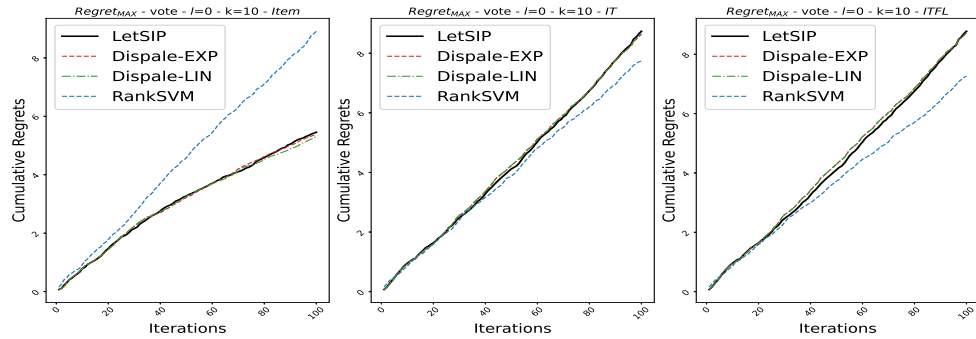
Fig. 4: A detailed view of experimental results w.r.t. $Regret_{\max}$, $k = 10$ and $\ell = 0$.



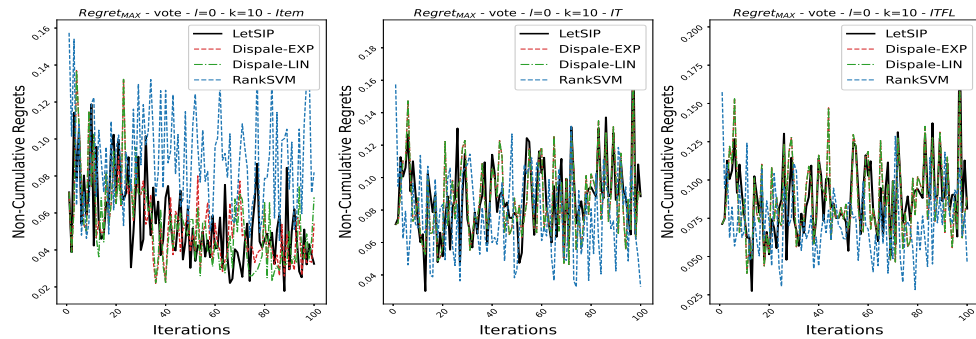
(a) Soybean: cumulative regret.



(b) Soybean: non cumulative regret.



(c) Vote: cumulative regret.



(d) Vote: non cumulative regret.

Fig. 5: A detailed view of experimental results w.r.t. $Regret_{\max}$, $k = 10$ and $\ell = 0$.

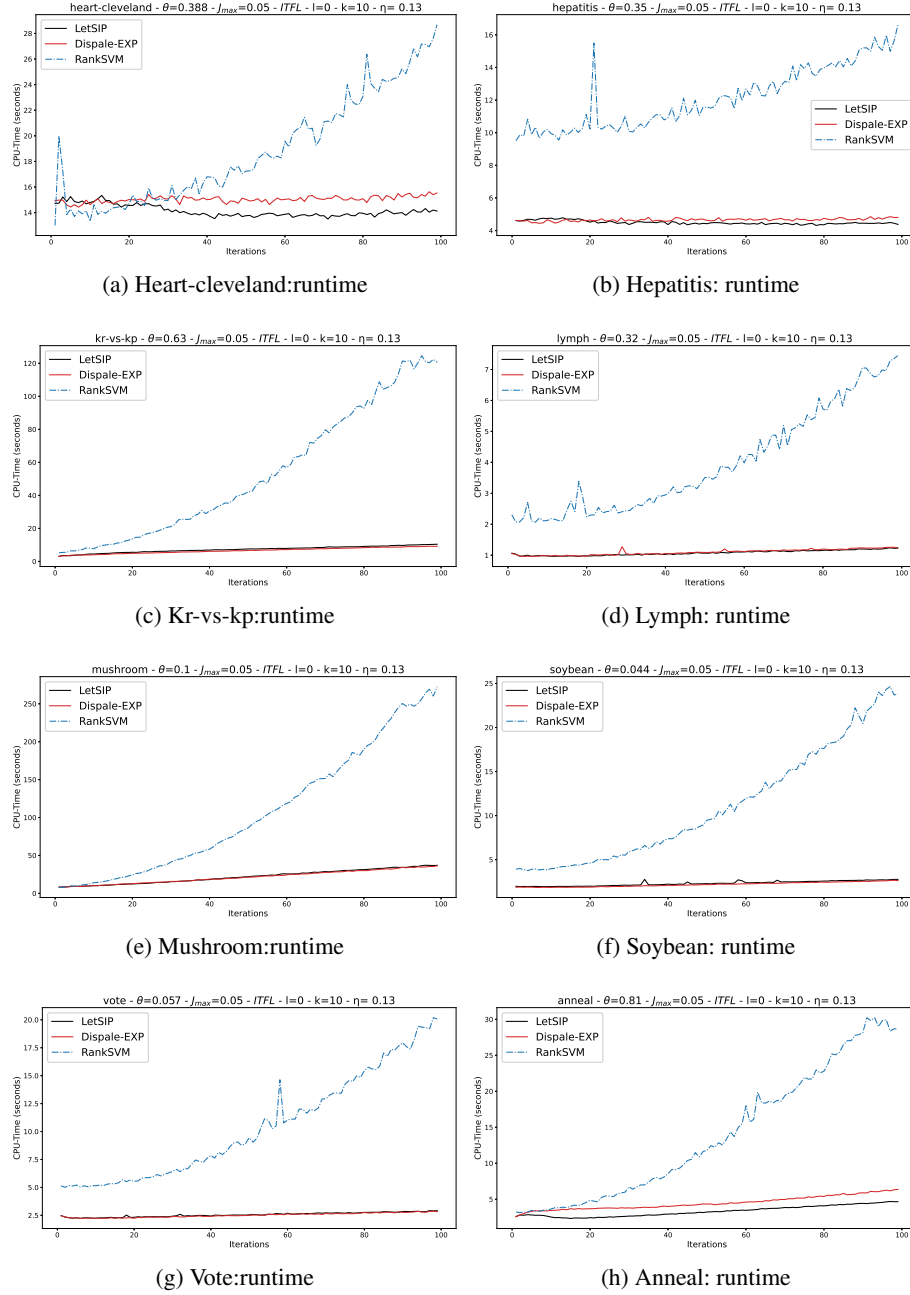


Fig. 6: Comparing the CPU-times of LETSIP, DISPALE and RANKSVM w.r.t. the ILFT feature combination with $k = 10$ and $\ell = 0$.

Table 1: Evaluation of feature I and comparison of DiSPALe ($\eta = 0.13$) with LETSIP and RANKSVM for $k = 10$. Results are given for each dataset. (1): LETSIP, (2): DiSPALe-EXP, (3): DiSPALe-LIN, (4): RANKSVM.

Datasets	$\ell = 0$							
	Regret: $Regret_{\max}$				Regret: $Regret_{Avg}$			
	(1)	(2)	(3)	(4)	(1)	(2)	(3)	(4)
anneal	8.799	9.141	9.141	9.057	49.200	49.674	49.674	49.860
chess	8.430	8.651	8.603	8.838	48.307	49.185	49.147	49.744
german-credit	8.197	8.152	8.120	8.921	47.955	46.937	46.954	49.631
heart-cleveland	8.389	8.021	8.028	8.987	47.916	46.801	46.829	49.109
hepatitis	8.400	8.265	8.265	8.276	47.694	46.544	46.544	48.999
kr-vs-kp	8.402	8.619	8.619	8.677	48.150	48.884	48.891	49.263
lymph	8.426	8.688	8.688	9.062	47.439	48.539	48.540	49.030
mushroom	34.389	34.638	34.546	35.150	87.800	87.570	87.580	88.295
soybean	7.567	8.974	8.976	9.170	47.061	48.194	48.192	49.649
vote	5.457	5.331	5.408	8.912	42.715	40.279	40.305	49.579
zoo-1	5.681	6.086	6.085	8.079	40.578	40.442	40.597	49.144

Datasets	$\ell = 1$							
	Regret: $Regret_{\max}$				Regret: $Regret_{Avg}$			
	(1)	(2)	(3)	(4)	(1)	(2)	(3)	(4)
anneal	0.512	0.513	0.513	0.467	44.869	44.927	44.927	43.824
chess	0.670	0.667	0.667	0.647	43.926	44.178	44.218	44.180
german-credit	0.518	0.506	0.504	0.627	42.605	41.598	41.802	44.422
heart-cleveland	0.171	0.346	0.346	0.482	44.182	43.206	43.206	44.391
hepatitis	0.554	0.647	0.647	0.521	40.116	41.005	41.005	43.962
kr-vs-kp	0.479	0.462	0.462	0.708	43.440	44.325	44.325	44.113
lymph	0.409	0.607	0.607	0.477	42.664	43.926	43.917	43.911
mushroom	5.856	6.625	6.625	5.964	79.223	80.043	80.043	79.924
soybean	0.489	0.374	0.369	0.531	41.818	43.175	42.914	43.956
vote	0.426	0.318	0.306	0.618	38.675	35.936	35.915	44.563
zoo-1	0.354	0.373	0.364	0.422	35.400	36.832	36.966	44.388

Table 2: Evaluation of features IT and comparison of DiSPaLE ($\eta = 0.13$) with LET-SIP and RANKSVM for $k = 10$. Results are given for each dataset. (1): LETSIP, (2): DiSPaLE-EXP, (3): DiSPaLE-LIN, (4): RANKSVM.

$\ell = 0$								
Datasets	Regret: $Regret_{\max}$				Regret: $Regret_{Avg}$			
	(1)	(2)	(3)	(4)	(1)	(2)	(3)	(4)
anneal	8.547	4.229	4.698	7.547	47.679	35.116	37.123	46.478
chess	4.790	3.068	3.040	5.338	44.293	37.243	37.428	41.485
german-credit	6.088	3.146	3.173	5.844	40.427	32.223	32.317	44.037
heart-cleveland	6.029	3.666	3.713	7.222	39.245	32.627	32.874	45.736
hepatitis	7.144	5.587	5.615	7.684	43.336	36.682	36.438	46.196
kr-vs-kp	5.508	2.760	2.673	5.536	44.648	36.402	36.360	41.326
lymph	8.329	8.351	8.347	7.270	48.756	47.625	47.639	46.403
mushroom	36.031	34.048	34.060	31.397	88.438	87.570	87.603	87.100
soybean	8.525	9.146	9.146	8.165	48.247	49.050	49.053	49.003
vote	8.736	8.646	8.646	7.742	49.608	49.592	49.590	47.174
zoo-1	8.720	8.881	8.867	7.889	48.880	48.836	48.822	47.657
$\ell = 1$								
Datasets	Regret: $Regret_{\max}$				Regret: $Regret_{Avg}$			
	(1)	(2)	(3)	(4)	(1)	(2)	(3)	(4)
anneal	0.367	0.275	0.281	0.577	42.784	32.320	32.189	42.087
chess	0.389	0.394	0.390	0.501	39.602	33.913	34.106	38.337
german-credit	0.378	0.248	0.248	0.438	35.377	29.589	29.678	40.145
heart-cleveland	0.081	0.084	0.084	0.460	38.629	27.503	27.994	40.878
hepatitis	0.365	0.324	0.324	0.500	36.716	31.035	31.077	42.063
kr-vs-kp	0.379	0.365	0.365	0.443	34.066	38.581	37.873	37.395
lymph	0.466	0.385	0.403	0.374	44.437	44.064	44.267	42.109
mushroom	6.859	7.829	7.829	4.393	80.392	79.831	79.840	78.540
soybean	0.609	0.605	0.605	0.347	43.805	44.187	44.188	43.970
vote	0.488	0.519	0.519	0.689	44.172	44.390	44.390	42.697
zoo-1	0.381	0.437	0.437	0.470	43.709	44.031	44.037	42.793

Table 3: Evaluation of features *ITLF* and comparison of DiSPaLE ($\eta = 0.13$) with LETSIP and RANKSVM for $k = 10$. Results are given for each dataset. (1): LETSIP, (2): DiSPaLE-EXP, (3): DiSPaLE-LIN, (4): RANKSVM.

$\ell = 0$								
Datasets	Regret: $Regret_{\max}$				Regret: $Regret_{Avg}$			
	(1)	(2)	(3)	(4)	(1)	(2)	(3)	(4)
anneal	8.634	3.624	4.504	7.556	47.650	32.793	34.781	46.779
chess	5.145	3.344	3.360	5.319	43.899	38.576	38.347	41.423
german-credit	5.668	2.956	3.008	6.077	40.230	31.533	31.593	43.331
heart-cleveland	5.700	3.718	3.572	7.047	39.081	31.564	31.790	45.479
hepatitis	7.138	5.398	5.253	7.669	43.150	36.410	36.058	46.080
kr-vs-kp	4.722	2.818	2.852	5.545	43.973	36.838	37.169	41.892
lymph	8.297	8.215	8.199	6.978	48.745	47.639	47.621	45.173
mushroom	36.364	33.205	33.205	30.884	88.603	87.549	87.549	87.074
soybean	8.508	9.264	9.264	7.976	47.866	49.052	49.054	48.921
vote	8.776	8.727	8.727	7.267	49.592	49.623	49.623	47.197
zoo-1	7.054	7.122	7.122	7.842	46.058	45.960	45.959	47.495
$\ell = 1$								
Datasets	Regret: $Regret_{\max}$				Regret: $Regret_{Avg}$			
	(1)	(2)	(3)	(4)	(1)	(2)	(3)	(4)
anneal	0.342	0.274	0.290	0.692	42.756	32.223	31.818	42.042
chess	0.400	0.378	0.377	0.384	39.335	34.014	33.812	37.609
german-credit	0.423	0.266	0.270	0.439	35.659	30.163	30.142	40.315
heart-cleveland	0.332	0.329	0.321	0.422	36.258	30.497	30.933	40.572
hepatitis	0.497	0.375	0.367	0.505	39.066	32.848	32.785	42.324
kr-vs-kp	0.349	0.310	0.309	0.460	40.921	33.724	34.037	38.272
lymph	0.512	0.420	0.408	0.424	44.458	44.149	44.131	41.944
mushroom	6.913	7.647	7.599	4.784	80.433	79.824	79.825	78.388
soybean	0.583	0.575	0.575	0.355	43.988	44.232	44.236	43.758
vote	0.495	0.555	0.555	0.645	44.157	44.546	44.546	42.762
zoo-1	0.429	0.449	0.449	0.489	43.787	43.980	43.973	42.664

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